## ANSWER KEY & MARKING SCHEME SET 1

Chapter 6: **How Forces Affect Motion** | Class 9 Science | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 2 | **(c)** 1 : 4 | a is inversely proportional to m. |
| 3 | **(d)** the pull of the book on the Earth | Third-law pair. |
| 4 | **(d)** 4 m/s squared | F/m. |
| 5 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 6 | **(d)** 20 m/s | a = 4 m/s squared; v = 20 m/s. |
| 7 | **(b)** reducing the friction by rollers | Reduce friction. |
| 8 | **(a)** 200 N | 0.1 x 20 / 0.01. |
| 9 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 10 | **(c)** 10 m/s squared downward | Only gravity acts. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | a = -2 m/s squared (1); t = 5 s, s = 25 m (1). | 1+1 |
| 12 | Normal force from the ground balances the weight (1); Net force zero (1). | 1+1 |
| 13 | Net force zero at constant velocity (1); Hence pull equals friction (1). | 1+1 |
| 14 | Same direction: 14/2 = 7 m/s squared (1); Opposite: 4/2 = 2 m/s squared (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | a = 4 m/s squared (1); F = 6000 N, s = 50 m (1); Weight = 15000 N; force is 0.4 of weight (1). | 1+1+1 |
| 16 | a = v squared/2s = 2 x 10^6 m/s squared (1); F = 0.01 x 2 x 10^6 = 20000 N (1); t = v/a = 0.0002 s (1). | 1+1+1 |
| 17 | F = 10/0.01 = 1000 N (1); Third law: nail exerts 1000 N on the hammer (1); Force on the nail 1000 N (1). | 1+1+1 |
| | **OR:** a = 5 m/s squared (1); v = 20 m/s, change in momentum 120 kg m/s (1); s = 40 m (1). | 1+1+1 |
| 18 | At 40 N: net force zero, constant velocity or rest (1); At 60 N: net 20 N (1); Box accelerates in the direction of the push (1). | 1+1+1 |
| | **OR:** To every action there is an equal and opposite reaction (1); Examples: recoil of a gun, rocket, walking (1); Action and reaction act on different bodies (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | a = 3 m/s squared (1); v = 15 m/s (1); s = 37.5 m (1); p = 120 kg m/s (1); F = change in p/t = 24 N (1). | 1+1+1+1+1 |
| | **OR:** Change = 0.16 x 70 = 11.2 kg m/s (1); Impulse = 11.2 N s (1); F = 560 N (1); Doubling time halves force (1); Same impulse (1). | 1+1+1+1+1 |
| 20 | Momentum before firing = 0 (1); Momentum of bullet = 0.02 x 200 = 4 kg m/s (1); Gun has equal momentum in the opposite direction (1); Recoil velocity = 4/4 = 1 m/s opposite to the bullet (1); Law of conservation of momentum (1). | 1+1+1+1+1 |
| | **OR:** Net = sqrt(900 + 1600) = 50 N (1); a = 10 m/s squared (1); tan = 40/30, about 53 degrees north of east (1); Second law (1); Vector addition (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

## ANSWER KEY & MARKING SCHEME SET 2

Chapter 6: **How Forces Affect Motion** | Class 9 Science | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** the same for both | F = change in p/t. |
| 2 | **(c)** 10 N | 20/2. |
| 3 | **(c)** A is true but R is false. | A is true; R is false: mass does not depend on gravity. |
| 4 | **(c)** 300 N | m a + f = 200 + 100. |
| 5 | **(b)** 1 : 2 | v = Ft/m. |
| 6 | **(b)** 10 N s | Ft. |
| 7 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 8 | **(c)** 6 N | a = 2; force on the 3 kg block = 6. |
| 9 | **(d)** 30 kg m/s | Impulse = Ft. |
| 10 | **(c)** 8 m/s squared | a = 2F/(m/2). |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | a = F/m = 40/8 (1); = 5 m/s squared (1). | 1+1 |
| 12 | Impulse 1 N s (1); v = 2 m/s (1). | 1+1 |
| 13 | F = rate of change of momentum (1); = 5 x 2000 = 10000 N (1). | 1+1 |
| 14 | Body keeps the forward velocity of the train (1); Feet stop on the ground, upper body continues and he falls (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | a = F/m = 60/12 = 5 m/s squared (1); v = at = 5 x 3 (1); = 15 m/s (1). | 1+1+1 |
| 16 | Limiting static friction = 250 N (1); Net force when moving = 100 N (1); a = 2 m/s squared (1). | 1+1+1 |
| 17 | a = 3 m/s squared (1); v = 15 m/s (1); From rest (1). | 1+1+1 |
| | **OR:** a = 5 m/s squared for 2 s: v = 10 m/s (1); Then constant 10 m/s (1); s = 10 + 30 = 40 m (1). | 1+1+1 |
| 18 | Same bullet momentum (1); Recoil speed inversely proportional to mass (1); Heavier gun recoils less (ratio 4 : 1) (1). | 1+1+1 |
| | **OR:** a = 2 m/s squared (1); v = 8 m/s (1); Change in momentum = 12 kg m/s (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Change in momentum 5 kg m/s (1); F1 = 100 N (1); F2 = 20 N (1); Same change, longer time gives smaller force (1); Used in cricket catching (1). | 1+1+1+1+1 |
| | **OR:** First law: ball at rest stays until force acts (1); Impulse = 100 x 0.05 = 5 N s (1); v = 5/0.4 = 12.5 m/s (1); Third law: the foot feels an equal and opposite force (1); Friction and air resistance finally stop the ball (1). | 1+1+1+1+1 |
| 20 | Change in momentum 900 kg m/s for passenger (1); With airbag (0.3 s): 3000 N (1); Without (0.1 s): 9000 N (1); Airbag reduces the force to one-third (1); F = change in momentum/time (1). | 1+1+1+1+1 |
| | **OR:** Engine force overcomes resistance (1); Net force gives acceleration (1); Inertia in braking (1); Seat belt, airbag, crumple zone (1); Speed limits reduce momentum (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

# THE MASTERMIND COACHING CLASSES

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## ANSWER KEY & MARKING SCHEME SET 3

Chapter 6: **How Forces Affect Motion** | Class 9 Science | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 2 | **(d)** moves away from the shore | Third law. |
| 3 | **(a)** 500 N | m a on the trailer. |
| 4 | **(a)** 4 m | v squared/2a. |
| 5 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 6 | **(d)** 7 m/s | a = 2; v = 3 + 4. |
| 7 | **(d)** a1 a2/(a1 + a2) | F/(m1 + m2) with m = F/a. |
| 8 | **(b)** 10 m/s | a = 2.5. |
| 9 | **(d)** A is false but R is true. | A is false: all bodies fall with the same acceleration; R is true but does not support A. |
| 10 | **(d)** 10 N | m a. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Friction at the axle and air drag (1); Unbalanced force slows it (1). | 1+1 |
| 12 | v = 10 m/s (1); F = 600/0.2 = 3000 N (1). | 1+1 |
| 13 | a = -2 m/s squared; F = 4 N (1); s = 9 m (1). | 1+1 |
| 14 | 100 N in the opposite direction (1); Wall does not move due to ground friction (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Air resistance increases with speed (1); At terminal velocity net force zero (1); Resistance = weight = 700 N (1). | 1+1+1 |
| 16 | Change = 0.15 x 70 = 10.5 kg m/s (1); F = 1050 N (1); Impulse 10.5 N s (1). | 1+1+1 |
| 17 | a1 = 6 m/s squared (1); a2 = 2 m/s squared (1); More inertia, less acceleration (1). | 1+1+1 |
| | **OR:** (a) 50 N (1); (b) 60 N (1); (c) 40 N (1). | 1+1+1 |
| 18 | The stone moves along the tangent (1); No force acts to change its direction (1); Inertia of motion (1). | 1+1+1 |
| | **OR:** F = 3 N (1); u = 4 m/s (1); v = 8 m/s (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Momentum per bullet 10 kg m/s (1); 10 bullets per s: 100 kg m/s per s (1); F = 100 N (1); Third law: the bullets push the gun back (1); Force to hold = 100 N (1). | 1+1+1+1+1 |
| | **OR:** a = 5/(2.5) = 2 m/s squared (1); For the weight: 5 - T = 0.5 x 2 (1); T = 4 N (1); Trolley: T = 2 x 2 = 4 N (1); Check consistent (1). | 1+1+1+1+1 |
| 20 | Net force 40 N (1); a = 4 m/s squared (1); v = 20 m/s (1); s = 50 m (1); p = 200 kg m/s (1). | 1+1+1+1+1 |
| | **OR:** a = 5 m/s squared (1); s = 40 m, t = 4 s (1); Double speed: s = 160 m, t = 8 s (1); Distance four times, time twice (1); Important for speed limits (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.